

Game Theory: Problem Set 4

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Problem 1. There are two players, a seller and a buyer. At date 1, the seller chooses his irreversible investment level $I \geq 0$ at cost I . At date 2, the seller may sell one unit of a good to the buyer, and the cost of supplying the good is $c(I)$ with $c' < 0$, $\lim_{I \rightarrow \infty} c'(I) = -\infty$, and $c'' > 0$. To simplify, assume $c(0)$ is less than the buyer's valuation for the good, v . There is no discounting.

- What is the socially optimal level of investment?
- Suppose that at date 2 the buyer observes the investment made by the seller I and makes him a take-it-or-leave-it offer to buy the good for p . What is this offer? What is the subgame-perfect Nash equilibrium of the game?
- Can you think of a contractual way of avoiding the inefficient outcome of part b? (Assume that contracts cannot be written on the level of I).

Part A.

Define the indicator

$$q = \begin{cases} 1 & \text{if good sold} \\ 0 & \text{otherwise} \end{cases}$$

Let $\pi(I, q) = pq - I - c(I)q$ be the seller's profit function and $CS = vq - pq$ be the buyer's "surplus." To find the socially optimal level of investment, the social planner wishes to maximize the total surplus function with respect to I .

$$\max_I TS(I, q) = \pi + CS = p \cdot q - I - c(I) \cdot q + v \cdot q - p \cdot q = v \cdot q - c(I) \cdot q - I$$

Taking the derivative of the objective function with respect to q yields

$$\frac{\partial TS}{\partial q} = v - c(I) > 0$$

since $v - c(0) > 0$ and $c(I)$ is decreasing in I . Define the function

$$q^*(p) = \begin{cases} 1 & \text{if } v \geq p \\ 0 & \text{otherwise} \end{cases}$$

$q^*(p) = 0 \implies TS(I, 0) = -I \leq 0$ and $q^*(p) = 1 \implies TS(I, 1) = v - c(I) - I$. To achieve the socially optimal level of investment, let $q^*(p) = 1$ since $I = 0$ is always possible, and $TS(I, 1) > 0$ when $I = 0$. The first-order condition of TS with respect to I is

$$c'(I) = -1$$

This implies that $I^* = c'^{-1}(-1)$ is the socially optimal level of investment.

Part B.

To determine the offer made by the buyer, as well as the subgame-perfect Nash equilibrium, we will solve the game using backward induction.

Third Stage

The seller maximizes his profit by playing a best response to the given price level and optimally choosing whether or not to sell the good.

$$\pi^*(q) = \begin{cases} p - I - c(I) & \text{if } q = 1 \\ -I & \text{if } q = 0 \end{cases}$$

and

$$q^*(p) = \begin{cases} 1 & \text{if } p \geq c(I) \\ 0 & \text{if } p < c(I) \end{cases}$$

Second Stage

The buyer selects a price for the good to maximize his “consumer surplus” given the best response function of the seller in the third stage.

$$\max_p CS(p) = v \cdot q^*(p) - p \cdot q^*(p)$$

There are two cases to consider:

$$p < c(I) \implies CS(p) = 0$$

and

$$p \geq c(I) \implies CS(p) = v - p$$

Since $\forall I : v > c(0) > c(I)$, then the buyer will maximize his consumer surplus by setting $p^*(I) = c(I) \implies \forall I : CS(p^*(I)) > 0$.

First stage

Given the best response functions of the previous stages, the seller maximizes his profit by optimally selecting a level of investment.

$$\max_I \pi(I) = p^*(I) \cdot q^*(p^*(I)) - I - c(I) \cdot q^*(p^*(I))$$

Since $p^*(I) = c(I)$, then $\pi(I) = -I$. Therefore, $I^* = 0$.

Therefore, the subgame-perfect Nash equilibrium is given by:

$$SPNE : (I^* = 0, q^* = 1; p^*(0) = c(0))$$

Part C.

To avoid the inefficient outcome of part (b), both the buyer and seller must agree to a price level in the first stage. To achieve the socially optimal level of investment, i.e., $I^* = c'^{-1}(-1)$, set $p^* = c(c'^{-1}(-1)) + c'^{-1}(-1)$

For the seller to agree to sell the good, $\pi(I) = p - c(I) - I \geq 0$. This implies that

$$\pi(I) = c(c'^{-1}(-1)) + c'^{-1}(-1) - c(c'^{-1}(-1)) - c'^{-1}(-1) = 0$$

So, the seller is indifferent between producing and not producing the good.

Now, for the buyer to agree to this price, $CS(p^*) > CS(c(0))$. Then,

$$v - (c(c'^{-1}(-1)) + c'^{-1}(-1)) > v - c(0)$$

which implies that

$$c(c'^{-1}(-1)) + c'^{-1}(-1) < c(0)$$

It is important to note that, $\forall I > 0 : TS(I, 1) > TS(0, 1)$. So,

$$\forall I > 0 : v - c(I) - I \geq v - c(0)$$

which implies that

$$\forall I > 0 : c(I) + I < c(0)$$

Therefore, the buyer will agree to the new price since it strictly improves his “surplus.”

Problem 2. Consider the following two stage model discussed in class. In the first stage, firms simultaneously choose whether to enter the market, resulting in a one time sunk cost of K for each entrant. In the second stage, the firms that chose to enter compete Cournot.

- Suppose that there are I identical consumers with individual demand functions $x_i = \alpha - \beta p$. Each firm has a constant marginal cost of production c . Derive the Cournot price and quantity as a function of the number of firms on the market J and the number of consumers I .
- Ignoring the integer problem, find the number of firms on the market that will choose to enter in the first stage in an SPNE of this game.
- What happens to the equilibrium number of entrants, the equilibrium price and quantity as $I \rightarrow \infty$? Explain.

Part A.

Endogenous entry: entry will occur as long as set-up costs K are less than potential profits. The totally quantity demanded is given by:

$$Q(p) = \sum_{i=0}^I (\alpha - \beta p) = I(\alpha - \beta p)$$

So, the inverse demand function is:

$$p(Q) = \frac{\alpha}{\beta} - \frac{Q}{I\beta}$$

Constant $MC = c > 0$. Timing: All potential firms simultaneously decide in or out where in is associated with a set-up cost K . All entrants J then play a Cournot game. We solve the game by backward induction.

Stage 2

Suppose that J firms have entered the market. Firm j 's optimization problem under Cournot is as follows:

$$\max_{q_j} (p(Q) - c) \cdot q_j = \left[\left(\frac{\alpha}{\beta} - \frac{1}{I\beta} \sum_{k=1}^J q_k \right) - c \right] \cdot q_j$$

The first-order condition is given by

$$\frac{\alpha}{\beta} - \frac{1}{I\beta} \left(2q_j + \sum_{k \neq j}^J q_k \right) - c = 0$$

So, the best response function for firm j is

$$r_j(q_{-j}) = \frac{IB}{2} \left(\frac{\alpha}{\beta} - \frac{1}{I\beta} \sum_{k \neq j}^J q_k - c \right)$$

We assume that we have a symmetric NE, i.e., $\forall j \neq k : q_j^* = q_k^* = q^*$. Now, we solve for q_j^* .

$$\begin{aligned} q_j^* &= \frac{IB}{2} \left(\frac{\alpha}{\beta} - \frac{(J-1)}{I\beta} q_j^* - c \right) \\ q_j^* + \frac{(J-1)}{2} q_j^* &= \frac{I\beta}{2} \left(\frac{\alpha}{\beta} - c \right) \\ \frac{2q_j^*}{2} + \frac{Jq_j^*}{2} - \frac{q_j^*}{2} &= \frac{I\beta}{2} \left(\frac{\alpha}{\beta} - c \right) \\ (J+1)q_j^* &= I\beta \left(\frac{\alpha}{\beta} - c \right) \\ q_j^* &= \frac{I\beta}{(J+1)} \left(\frac{\alpha}{\beta} - c \right) \\ &= \frac{I\alpha}{(J+1)} - \frac{I\beta c}{(J+1)} \end{aligned}$$

Thus, we can derive the Cournot price and quantity

$$Q^* = \sum_{j=1}^J q_j^* = \frac{JI\beta}{(J+1)} \left(\frac{\alpha}{\beta} - c \right) = \frac{JI(a - \beta c)}{(J+1)}$$

$$p^*(Q^*) = \frac{\alpha}{\beta} - \frac{J}{(J+1)} \left(\frac{\alpha}{\beta} - c \right) = \frac{\alpha}{\beta} - \frac{J}{(J+1)} \frac{\alpha}{\beta} + \frac{J}{(J+1)} c$$

Part B.

Ignoring the integer problem, we find the number of firms on the market that will choose to enter in the first stage in an SPNE of this game.

$$\begin{aligned} \pi_j^* &= p(Q^*) \cdot q_j^* - c \cdot q_j^* \\ &= \left[\frac{\alpha}{\beta} - \frac{J}{(J+1)} \left(\frac{\alpha}{\beta} - c \right) \right] \left[\frac{I\beta}{(J+1)} \left(\frac{\alpha}{\beta} - c \right) \right] - c \left[\frac{I\beta}{(J+1)} \left(\frac{\alpha}{\beta} - c \right) \right] \\ &= \frac{I}{(J+1)} \frac{\alpha^2}{\beta} - \frac{J}{(J+1)} \frac{I}{(J+1)} \frac{\alpha^2}{\beta} + \frac{J}{(J+1)} \frac{2I}{(J+1)} \alpha c \\ &\quad - \frac{2I}{(J+1)} \alpha c + \frac{I\beta}{(J+1)} c^2 - \frac{J}{(J+1)} \frac{I\beta}{(J+1)} c^2 \\ &= \left[1 - \frac{J}{(J+1)} \right] \frac{I}{(J+1)} \frac{\alpha^2}{\beta} + \left[\frac{J}{(J+1)} - 1 \right] \frac{2I}{(J+1)} \alpha c + \left[1 - \frac{J}{(J+1)} \right] \frac{I\beta}{(J+1)} c^2 \end{aligned}$$

Stage 1: Entry

Let J^* be the the SPNE entrants in the market. No deviation incentives requires: $\pi^*(J^*) \geq K$ and $\pi^*(J^* + 1) < K$. Firm J is exactly indifferent between entry and staying out if $\pi_j^* = K$.

$$\begin{aligned} K &= \left[1 - \frac{J}{(J+1)} \right] \frac{I}{(J+1)} \frac{\alpha^2}{\beta} \\ &\quad + \left[\frac{J}{(J+1)} - 1 \right] \frac{2I}{(J+1)} \alpha c + \left[1 - \frac{J}{(J+1)} \right] \frac{I\beta}{(J+1)} c^2 \\ K(J+1)^2 &= [(J+1) - J] I \frac{\alpha^2}{\beta} \\ &\quad + [J - (J+1)] 2I\alpha c + [(J+1) - J] I\beta c^2 \\ K(J+1)^2 &= \frac{I\alpha^2}{\beta} - 2I\alpha c + I\beta c^2 \\ J^* &= \sqrt{\frac{I\alpha^2}{K\beta} - \frac{2I\alpha c}{K} + \frac{I\beta c^2}{K}} - 1 \\ &= (a - c\beta) \sqrt{\frac{I}{K\beta}} - 1 \end{aligned}$$

Part C.

What happens to the equilibrium number of entrants, the equilibrium price and quantity as $I \rightarrow \infty$?

$$J^* = (a - c\beta) \sqrt{\frac{I}{K\beta}} - 1$$

As $I \rightarrow \infty$, the equilibrium number of entrants approaches infinity, i.e., $J^* \rightarrow \infty$.

$$p^* = \frac{\alpha}{\beta} - \frac{J}{(J+1)} \frac{\alpha}{\beta} + \frac{J}{(J+1)} c$$

Since $J^* \rightarrow \infty$, we apply L'hospital's Rule to the equation for p^* to find that the equilibrium price approaches constant marginal cost as $I \rightarrow \infty$, i.e., $p^* \rightarrow c$.

$$Q^*(p^*) = \sum_{i=0}^I (\alpha - \beta p^*) = I(\alpha - \beta p^*)$$

At equilibrium, quantity demanded equals quantity supplied, so we can easily derive what happens to equilibrium quantity as $I \rightarrow \infty$ by analyzing aggregate quantity demand as a function of price. It was already determined that $p^* \rightarrow c$, so $(\alpha - \beta p^*) \rightarrow \text{constant}$ as $I \rightarrow \infty$. Hence, $Q^*(p^*) \rightarrow \infty$ as $I \rightarrow \infty$.

Problem 3. Consider the following matrix game Γ :

		PL 2	
		Y_2	Z_2
PL 1	Y_1	0, 0	-1, 2
	Z_1	2, -1	0, 0

- Does either player have a strictly dominant strategy in Γ ? If so, what is it?
- Find all Nash equilibria (mixed and pure) of Γ .
- Now, consider the super game $\Gamma^\infty(\delta)$, which consists of repeatedly playing Γ forever and in which the players possess the common discount factor $\delta \in (0, 1)$. Suppose that the players wish to support a SPNE in which they play (Y_1, Z_2) in even period ($t = 0, 2, 4, \dots$) and (Z_1, Y_2) in odd periods ($t = 1, 3, 5, \dots$). What is the discounted date-zero payoff to the players on the path suggested above?
- Find the value $\underline{\delta}$ such that the path suggested above can be supported in SPNE if and only if $\delta \geq \underline{\delta}$.

Part A.

Z_1 strictly dominates Y_1 and Z_2 strictly dominates Y_2 .

Part B.

By performing iterated elimination of strictly dominated strategies, we achieve the unique pure strategy Nash equilibrium: (Z_1, Z_2) . There is not a mixed strategy Nash equilibrium because neither player would put positive probability on their strictly dominated strategies.

Part C.

Now consider the supgame $\Gamma^\infty(\delta)$, which consists of repeatedly playing Γ forever and in which the players possess the common discount factor $\delta \in (0, 1)$. Suppose that the players wish to support a SPNE in which they play (Y_1, Z_2) in even periods ($t = 0, 2, 4, \dots$) and (Z_1, Y_2) in odd periods ($t = 1, 3, 5, \dots$). What is the discounted date-zero payoff to the players on the path suggested above?

$$\begin{aligned} u_1 &= \sum_{t=0}^{\infty} (-1)\delta^{2t} + 2\delta \sum_{t=0}^{\infty} \delta^{2t} \\ &= (2\delta - 1) \sum_{t=0}^{\infty} \delta^{2t} \end{aligned}$$

and

$$\begin{aligned} u_2 &= 2 \sum_{t=0}^{\infty} \delta^{2t} - \delta \sum_{t=0}^{\infty} \delta^{2t} \\ &= (2 - \delta) \sum_{t=0}^{\infty} \delta^{2t} \end{aligned}$$

Part D.

Find the value δ_* such that the path suggested above can be supported in SPNE if and only if $\delta \geq \delta_*$.

$$\begin{aligned} u_1 &= \frac{2\delta - 1}{1 - \delta^2} \geq 0 \implies \delta \geq \frac{1}{2} \\ u_2 &= \frac{2 - \delta}{1 - \delta^2} \geq 0 \implies 2 \geq \delta \\ &\implies \delta_* = \frac{1}{2} \end{aligned}$$

Problem 4. Suppose that there are two identical firms serving the market with inverse demand of

$$p = 1 - Q$$

where Q denotes the market output and p denotes the market price. Each firm's cost function is $C_i(q_i) = 0$ for $i = 1, 2$.

- Consider a quantity-setting supgame. Calculate the lowest discount parameter $\underline{\delta}_C$ such that collusion on the monopoly output, in which the two firms share the earnings evenly can be supported in a SPNE with a trigger strategy of Cournot-Nash reversion.

- b. Consider a symmetric price setting supergame. Calculate the lowest discount parameter $\underline{\delta}_B$ such that collusion on the monopoly price, in which the two firms share the earnings evenly can be supported in a SPNE with a trigger strategy of Bertrand-Nash reversion.

Part A.

The monopoly problem can be given as

$$\max_Q (1 - Q)Q$$

The first-order condition gives

$$\begin{aligned} 1 - 2Q &= 0 \\ Q^m &= \frac{1}{2} \end{aligned}$$

So,

$$\begin{aligned} p^m &= \frac{1}{2} \\ q_i^m &= \frac{1}{4} \\ \pi_i^m &= \frac{1}{8} \end{aligned}$$

The Cournot-Nash problem for firm i can be written as:

$$\max_{q_i} (1 - q_i - q_j)q_i$$

The first-order condition gives

$$\begin{aligned} 1 - 2q_i - q_j &= 0 \\ q_i^c &= \frac{1 - q_j^c}{2} \end{aligned}$$

Thus, we have

$$\begin{aligned} q_i^c &= \frac{1}{2} - \frac{1 - q_i^c}{4} \\ &= \frac{1}{4} + \frac{q_i^c}{4} \\ q_i^c &= \frac{1}{3} \\ p_i^c &= \frac{1}{3} \\ \pi_i^c &= \frac{1}{9} \end{aligned}$$

If one of the firms chooses to deviate, then its optimal q_i would be

$$\frac{1 - 1/4}{2} = \frac{3}{8}$$

So, $Q = 5/8$, $p = 3/8$, and thus, $\pi_i = 9/64$ and $\pi_j = 3/32$. If both firms collude, then each firm's discounted payoff is

$$EU_i = \frac{1}{8}(1 + \delta + \delta^2 + \dots) = \frac{1}{8(1 - \delta)}$$

Firm i 's payoff from deviating at $t = 0$ is

$$\begin{aligned} EU_i &= \frac{9}{64} + \frac{1}{9}(\delta + \delta^2 + \dots) \\ &= \frac{9}{64} + \frac{\delta}{9(1 - \delta)} \end{aligned}$$

In order for collusion to be supported, we require:

$$\begin{aligned} \frac{1}{8(1 - \delta)} &\geq \frac{9}{64} + \frac{\delta}{9(1 - \delta)} \\ 1 &\geq \frac{9}{8} - \frac{9\delta}{8} + \frac{8\delta}{9} \\ \delta &\geq \frac{9}{17} \end{aligned}$$

So, the lowest discount factor for which collusion can be supported is $\underline{\delta}_C = 9/17$

Part B.

The monopoly problem can be written as

$$\max_p p(1 - p)$$

The first-order condition gives

$$\begin{aligned} 1 - 2p &= 0 \\ p^m &= \frac{1}{2} \\ Q^m &= \frac{1}{2} \\ q_i^m &= \frac{1}{2} \\ \pi_i^m &= \frac{1}{8} \end{aligned}$$

If one of the firm's deviates from the monopoly price level, then that firm would get the entire profit share. So, the firm deviating from collusion would choose p_i slightly less than $1/2$, and thus,

their profit will be slightly less than $1/4$. For the Bertrand-Nash problem, we have $p_i^b = p_j^b = 0$ because each firm has a cost of zero. So, in this case, profit for each firm will also be zero. Thus, the payoff of firm i from collusion will be

$$EU_i = \frac{1}{8}(1 + \delta + \delta^2 + \dots)$$

$$EU_i = \frac{1}{8(1 - \delta)}$$

Deviating from collusion in the first period would yield

$$EU_i < \frac{1}{4}$$

In order for collusion to be supported, we should have

$$\frac{1}{8(1 - \delta)} \geq \frac{1}{4}$$

$$\delta \geq \frac{1}{2}$$

Therefore, the lowest discount parameter to support collusion is $\underline{\delta}_B$.